



Tecnológico Nacional de México

Instituto Tecnológico de Durango

Lenguajes y Automatas I

Ingeniería en Sistemas Computacionales

Unidad 5

First & Follows LIA

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First & Follows LIA

AI

FIRST

PROGRAM $\Rightarrow \{ \text{class} \}$

DECLARAR $\Rightarrow \{ \text{def}, \epsilon \}$

IDN $\Rightarrow \{ \text{g}, \epsilon \}$

TIPO $\Rightarrow \{ \text{int}, \text{float}, \text{char}, \text{bool}, \text{string} \}$

ESTATUTOS $= \{ \text{FIRST}(\text{EST_ASIG}), \dots, \epsilon \}$

ESTATUTOS $= \{ \text{id}, \text{if}, \text{while}, \text{do}, \text{input}, \text{output}, \text{loop}, \text{break}, \epsilon \}$

EST_ASIG $= \{ \text{id} \}$

EST_WRITE $= \{ \text{input} \}$

EST_READ $= \{ \text{output} \}$

EST_BREAK $= \{ \text{break} \}$

EST_DO $= \{ \text{do} \}$

EST_IF $= \{ \text{if} \}$

EST_WHILE $= \{ \text{while} \}$

EST_LOOP $= \{ \text{loop} \}$

ELSES $= \{ \text{else}, \epsilon \}$

EXPRN $= \{ \text{,}, \epsilon \}$

EXPR $= \{ \text{FIRST}(\text{EXPR2}) \} = \{ \text{!, id, cteentera, cterreal}, \dots, \epsilon \}$

EXPROR $= \{ \text{||}, \epsilon \}$

EXPR2 $= \{ \text{FIRST}(\text{EXPR3}) \} = \{ \text{!, id, cteentera, cterreal}, \dots, \epsilon \}$

EXPRAND $= \{ \&\&, \epsilon \}$

EXPR3 $= \{ \text{FIRST}(\text{NOT}) \} = \{ \text{!, FIRST}(\text{EXPR4}) \} = \{ \text{!, id, cteentera}, \dots, \epsilon \}$

NOT $= \{ \text{!, } \epsilon \}$

EXPR4 $= \{ \text{FIRST}(\text{EXPR5}) \} = \{ \text{id, cteentera, cterreal}, \dots, \epsilon \}$

EXPRREL $= \{ \text{FIRST}(\text{OPREL}), \epsilon \} = \{ \text{=, !=, <, <=, >, >=}, \epsilon \}$

OPREL $= \{ \text{=, !=, <, <=, >, >=} \}$

EXPR5 $= \{ \text{FIRST}(\text{TERM}) \} = \{ \text{id, cteentera, cterreal}, \dots, \epsilon \}$

ADI $= \{ \text{+, -}, \epsilon \}$

TERM $= \{ \text{FIRST}(\text{FACT}) \} = \{ \text{id, cteentera}, \dots, \epsilon \}$

PROD $= \{ \text{*, /, \%}, \epsilon \}$

FACT $= \{ \text{id, cteentera, cterreal, ctenotacion, ctecaracter, ctestring}, \epsilon \}$

FIRST & FOLLOWS ALIA

FOLLOWS

PROGRAM = { \$ }

DECLARAR = { FIRST(ESTATUTOS), FOLLOWS(DECLARAR) } = { id, if, while, do, input, output, loop, break, ε }

IDN = { :, FOLLOWS(IDN), } = { :, }

TIPO = { ; }

ESTATUTOS = { }, FOLLOWS(ESTATUTOS), do while, FIRST(ELSE), FOLLOW(ELSE) } = { }, do while, else, FOLLOW(ELSE) } = { }, do while, else }

EST_ASSIGN = { FIRST(ESTATUTOS) } = { id, if, while, do, input, output, loop, break, ε }

EST_WRITE = { FIRST(ESTATUTOS) } = { id, if, while, do, input, output, loop, break, ε }

EST_READ = { FIRST(ESTATUTOS) } = { id, if, while, do, ..., ε }

EST_BREAK = { FIRST(ESTATUTOS) } = { id, if, while, do, ..., ε }

EST_DO = { FIRST(ESTATUTOS) } = { id, if, while, do, input, ..., ε }

EST_IF = { FIRST(ESTATUTOS) } = { id, if, while, do, input, ..., ε }

ELSE = { }

EST_WHILE = { FIRST(ESTATUTOS) } = { id, if, while, do, input, ..., ε }

EST_LOOP = { FIRST(ESTATUTOS) } = { id, if, while, do, input, ..., ε }

EXPRN = { }, FOLLOWS(EXPRN) } = { }

EXPR = { ;, FIRST(EXPRN),), FOLLOWS(EXPRN) } = { ;, , FOLLOWS(EXPRN),), FOLLOWS(EXPRN) } = { ;, ,), FOLLOWS(EXPRN) } = { ;, ,) FOLLOWS(EXPRN) }

EXPROR = { FOLLOWS(EXPR) } = { ;, ,) }

EXPR2 = { FIRST(EXPROR), FOLLOWS(EXPRAND) } = { ||, FOLLOWS(EXPROR), FOLLOWS(EXPR2) } = { ||, ;, ,) }

EXPRAND = { FOLLOWS(EXPR2) } = { ||, ;, ,) }

EXPR3 = { FIRST(EXPRAND) } = { &, FOLLOWS(EXPRAND) } = { &, ||, ;, ,) }

NOT = { FIRST(EXPR4) } = { id, cteentera, cte real, ctenotacion, ..., }

EXPR4 = { FOLLOWS(EXPR3) } = { &, ||, ;, ,) }

EXPRREL = { FOLLOWS(EXPR4) } = { &, ||, ;, ,) }

OPREL = { FIRST(EXPR5) } = { id, cteentera, cte real, ..., }

EXPR5 = { FIRST(EXPRREL), FOLLOWS(EXPRREL), FOLLOWS(ADI) } = { =, !=, <, <=, >, >=, &, ||, ;, ,), FOLLOWS(EXPR5) } = { =, !=, <, <=, >, >=, ..., }

FIRST And Follows - LIA

FOLLOWS

ADI = { FOLLOWS(EXPR) } = { =, !=, <, <=, >, >=, &&, ||, !, , ,) }

TERM = { FIRST(ADI), FOLLOWS(PROD) } = { +, -, FOLLOWS(ADI), FOLLOWS(TERM) }
= { +, -, =, !=, <, <=, >, >=, &&, ||, !, , ,) }

PROD = { FOLLOWS(TERM) } = { +, -, =, !=, <, <=, >, >=, , , ,) }

FACT = { FIRST(PROD) } = { *, /, %, FOLLOWS(PROD) } = { *, /, %, +, -, =, !=, <, <=, >, >=, &&, ||, !, , ,) }